INTEGRALS LEADING TO LOGARITHMS

The secant integral

Nothing in sec x suggests a logarithm. Multiplying by a clever form of 1 creates the f′/f pattern.

01

NO PATTERN YET — BEFORE

$$\int \sec x \, dx$$

↓ Multiply by a form of 1 — (sec x + tan x)/(sec x + tan x)

02

NUMERATOR IS THE DERIVATIVE OF THE DENOMINATOR

$$\int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx$$

↓ Integrate f′/f — ∫ f′/f dx = ln|f| + C, with f = sec x + tan x

03

LOGARITHM — AFTER

$$\ln \left| \sec x + \tan x \right| + C$$